Quantum Neural Networks

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Quantum Computing

Quantum computing is a revolutionary type of computation that leverages the principles of quantum mechanics to solve complex problems that are intractable for classical computers. It utilizes quantum bits (qubits) instead of classical bits, which differs from bits due to 4 major principles[1]:

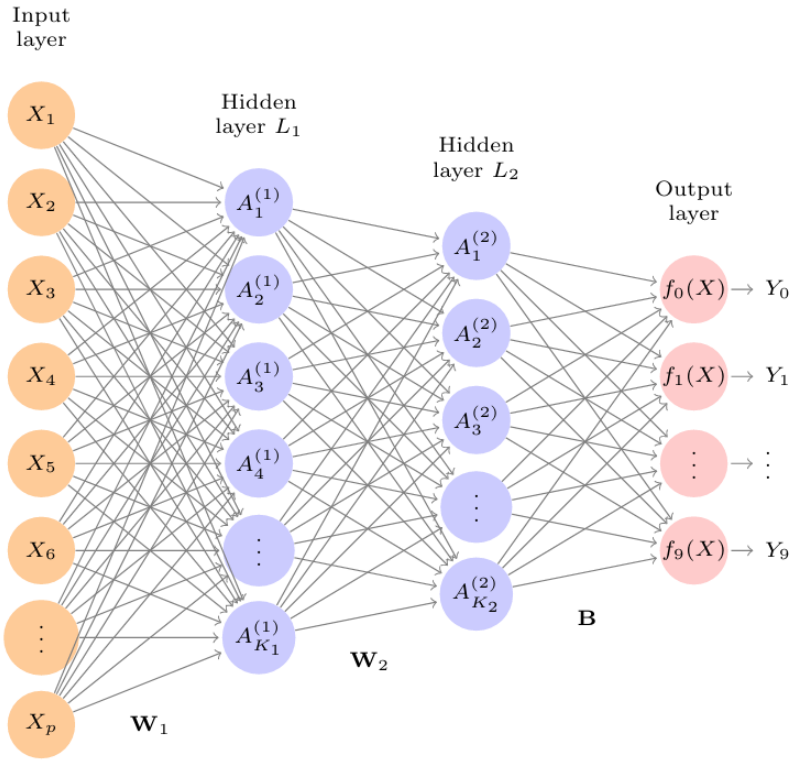
- Superposition: Quantum mechanics also permits the system to exist in multiple states simultaneously. This phenomenon is called quantum superposition and can be expressed as a linear combination of two states $a|0> + b|1>$.
- Entanglement: Entanglement is a multi-particle state that cannot be expressed as the product of two single-particle states. if the system is in a state such as $|01>+ |10>$, the measurement results for A and B are strongly correlated, and the two particles are entangled.
- Interference: Superposition also results in interference effects, where the probability amplitudes of different states can reinforce or cancel each other out after the state evolution.
- Decoherence :Decoherence is the process in which a system in a quantum state collapses into a nonquantum state. It can be intentionally triggered by measuring a quantum system or by other environmental factors.

Neural Networks

Neural networks are a type of machine learning algorithm that, inspired by the structure and function of the human brain, are used to process data and learn from it.

A neural network takes an input vector of p variables $X = (X_1, X_2, \ldots, X_p)$ and builds a nonlinear function f(X) to predict the response Y as shown in the Figure [2]

It contains Three major layers : Input Layer, Hidden Layer, Output Layer

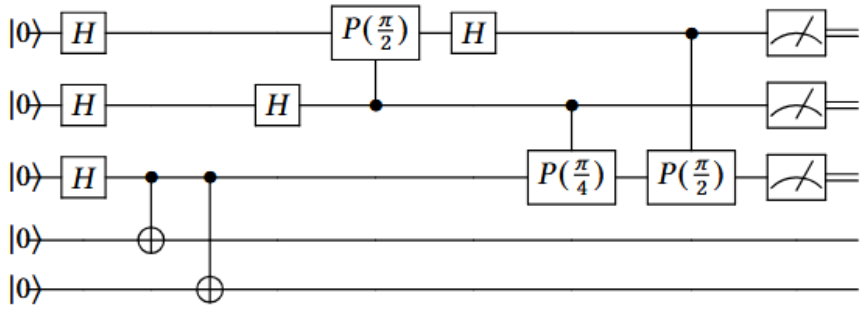


Quantum Algorithms

A quantum algorithm consists of three basic steps[3]:

- Encoding of the data, which could be classical or quantum, into the state of a set of input qubits.
- A sequence of quantum gates applied to this set of input qubits.
- Measurements of one or more of the qubits at the end to obtain a classically interpretable result.

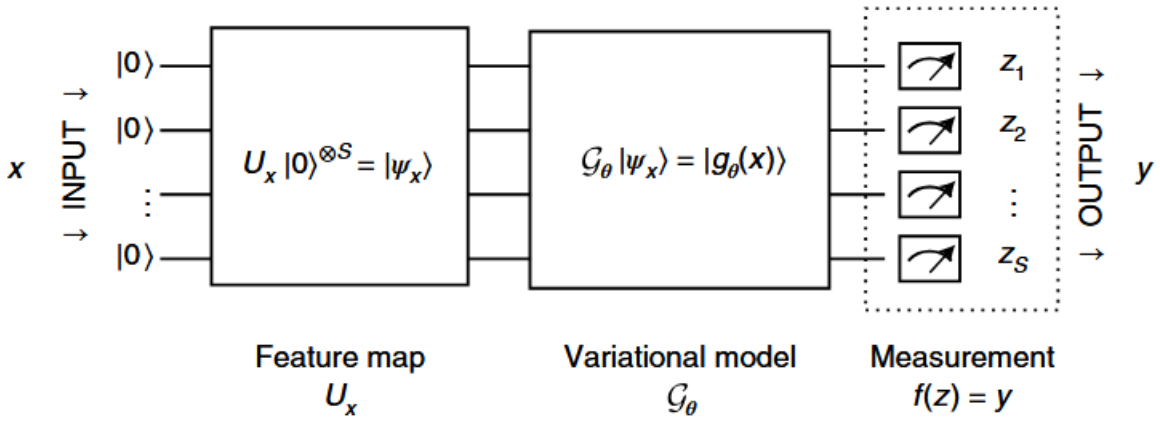
Below is a famous example of Shor's Algorithm.



Quantum Neural Networks

- Quantum neural networks are a sub-class of variational quantum algorithms that comprise quantum circuits containing parameterized gate operations (usually in the form of classical data) is first encoded into a quantum state via a state-preparation routine called a quantum feature map. [4]

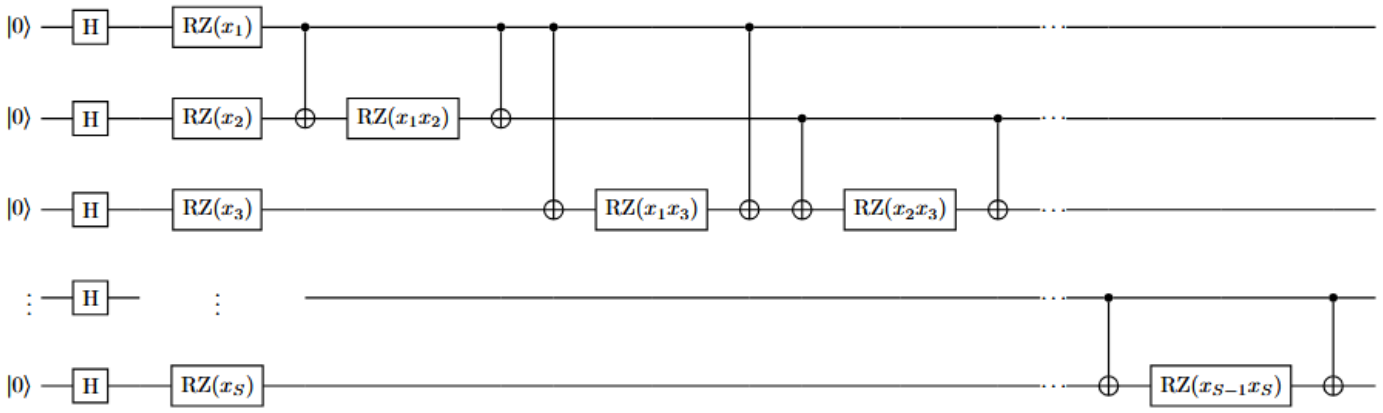
- Once data are encoded into a quantum state, a model called a variational model is applied, which contains parameterized gate operations that are optimized for a particular task, analogous to classical machine learning techniques. The final output of the quantum neural network is extracted from measurements made to the quantum circuit after the variational model is applied. These measurements are often converted to labels or predictions through classical postprocessing before being passed to a loss function, where the idea is to choose parameters for the variational model that minimize the loss function.



Basic Diagram Of Quantum Neural Network

Feature Mapping

A quantum feature map is a function that transforms classical data into a quantum state within a Hilbert space. This process allows quantum computers to analyze data in a way that leverages the unique properties of quantum mechanics, such as superposition and entanglement, potentially leading to enhanced machine learning capabilities[6]. Given below is the feature mapping of Quantum Neural Network



First, Hadamard gates are applied to each qubit. Then, normalized feature values of the data are encoded using RZ-gates. This is followed by CNOT-gates and higher order data encoding between every pair of qubits, and every pair of features in the data. The feature map is repeated to create a depth of 2. The easy quantum model, introduced in the results section, applies only the first sets of Hadamard and RZ-gates.[4]

Fisher Information Matrix

A way to assess the information gained by a particular parameterization of a statistical model is epitomized by the Fisher information. By defining a neural network as a statistical model, we can describe the joint relationship between data pairs

$$((x, y))\ as\ p(x, y;\ \theta) =\ p(y|x;\theta)p(x)\ \ x\epsilon\boldsymbol{X}\ y\epsilon\boldsymbol{Y}\ (\theta \in \Theta \subset [-1,1]^d)$$

where θ is a vectorized parameter set, Θ is the full parameter space and d is the number of trainable parameters.
This is achieved by applying an appropriate post-processing function. In the classical network, we apply a softmax function to the final layer, whereas in the quantum network we obtain probabilities based on the parity of the output bit strings. The input distribution p(x) is a prior distribution, whereas the conditional distribution p(y|x;θ) describes the input–output relation of the model for a fixed θ ∈ Θ; Θ forms a Riemannian space, which gives rise to a Riemannian metric, namely, the Fisher information matrix

$$F(\theta) = \mathbb{E}_{(x,y)\sim p}\left[\ \frac{\partial}{\partial\theta}\log p(x, y;\theta)\frac{\partial}{\partial\theta}\log p(x, y;\theta)^{\mathrm{T}}\ \right] \in \mathbb{R}^{d\times d},$$

the Fisher information matrix is positive semidefinite and hence its eigenvalues are non-negative, real numbers. The Fisher information conveniently helps capture the sensitivity of a neural network's output relative to movements in the parameter space.

Effective Dimension

- The effective dimension is a complexity measure motivated by information geometry, with useful qualities. The goal of the effective dimension is to estimate the size that a model occupies in model space—the space of all possible functions for a particular model class, where the Fisher information matrix serves as the metric.
- The effective dimension of a statistical model
- $M_\Theta := \{p(\cdot,\cdot;\theta)\colon \theta \in \Theta\}$ with respect to $\gamma\epsilon(0,1]$ a d-dimensional parameter space $\Theta \subset R^d$ and $n \in N, n > 1$ data samples is defined as

$$d_{\gamma,n}(\mathcal{M}_\Theta) := 2\frac{\log\left(\frac{1}{V_\Theta}\int_\Theta \sqrt{\det\left(\mathrm{id}_d + \frac{\gamma n}{2\pi\log n}\hat{F}(\theta)\right)}\,\mathrm{d}\theta\right)}{\log\left(\frac{\gamma n}{2\pi\log n}\right)},$$

- where $V_\Theta := \int d\theta \in R_+$ is the volume of the parameter space. The matrix $\hat{F}(\theta) \in R^{\{d\times d\}}$ is the normalized Fisher information matrix defined as

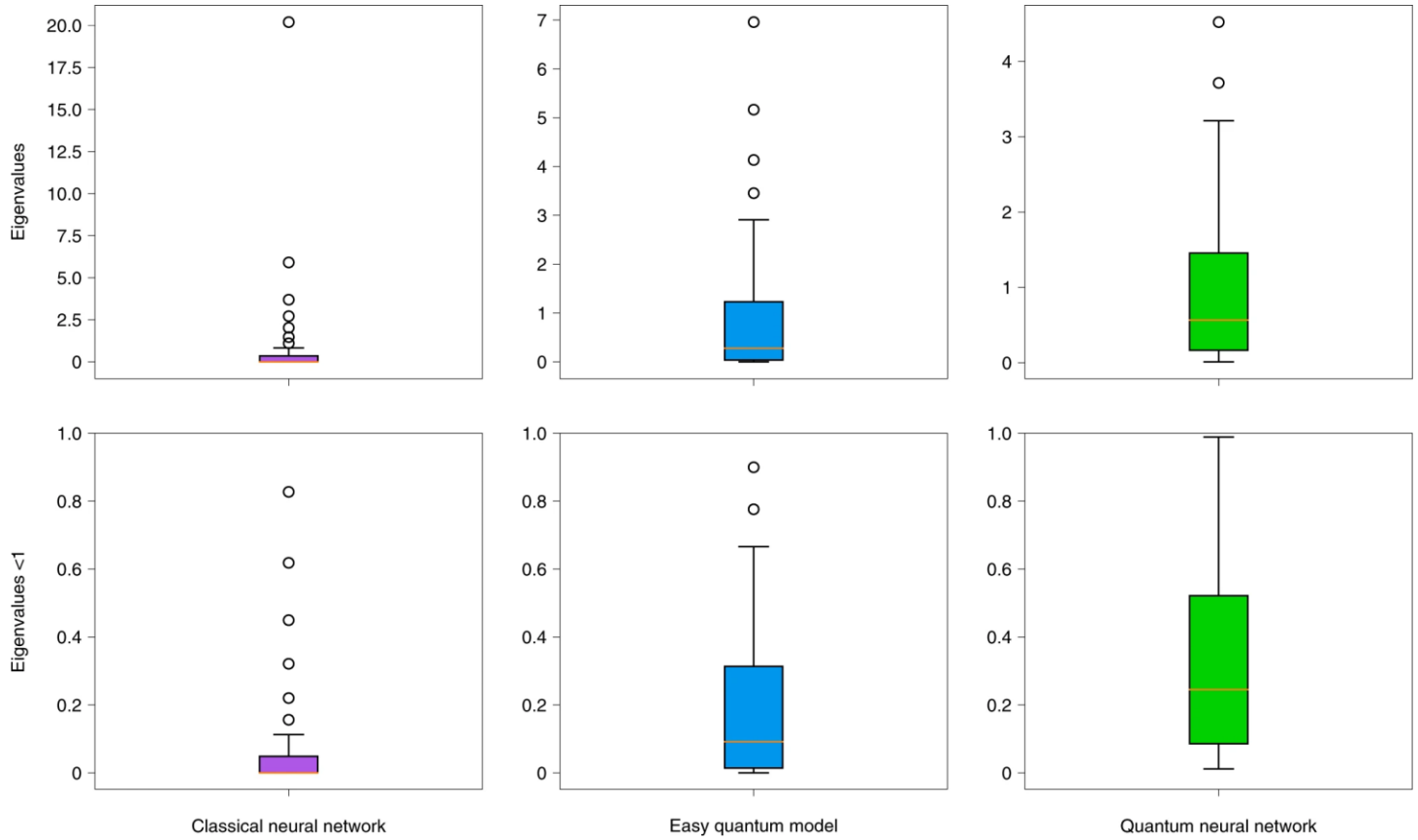
$$\hat{F}_{ij}(\theta) := d\frac{V_\Theta}{\int_\Theta \mathrm{tr}(F(\theta))\mathrm{d}\theta}F_{ij}(\theta)\,.$$

In the limit $n \to \infty$, the effective dimension converges to the maximal rank but it is worthwhile to note that the effective dimension does not necessarily increase monotonically with n.

Fisher Information Spectrum

- Classically, the Fisher information spectrum reveals a lot about the optimization landscape of a model. The magnitude of the eigenvalues illustrates the curvature of a model for a particular parameterization. If there is a large concentration of eigenvalues near zero, the optimization landscape will be predominantly flat and parameters become difficult to train with gradient-based methods
- On the quantum side if a model is in a barren plateau, the Fisher information spectrum will be concentrated around zero and training also becomes unfeasible. We can thus make connections to trainability via the spectrum of the Fisher information matrix by using the effective dimension.
- When the cost function exhibits a barren plateau, the cost function gradient vanishes exponentially with the system size.[5]

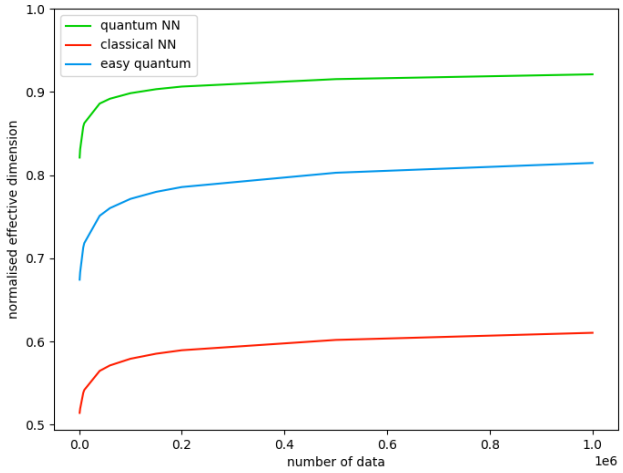
Fisher Information Spectrum

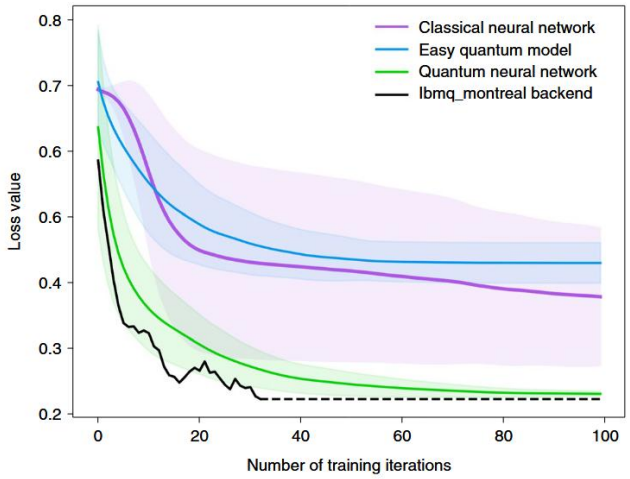


Box Plots of Average Eigenvalue Distribution of Fisher Information Matrix with Classical Neural Network And Two different quantum encodings.[4]

Trainability

The observed Fisher information spectrum of the feedforward model is known to have undesirable optimization properties, where the outlying eigenvalues slow down training and loss convergence. These large eigenvalues become even more pronounced in bigger models as seen in previous figure.
On examining the easy quantum model over an increasing system size,the average Fisher spectrum becomes more concentrated around zero. This is characteristic of models encountering a barren plateau, presenting another unfavorable scenario for optimization. The quantum neural network, however, maintains its more even distribution of eigenvalues as the number of qubits and trainable parameters increase. Furthermore, a large amount of the eigenvalues are not near zero.
This highlights the importance of a feature map in a quantum model. The harder data encoding strategy used in the quantum neural network seems to structurally change the optimization landscape and remove the flatness, usually associated with suboptimal optimization conditions such as barren plateaus.





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Summary and Future works

- ✔ In stark contrast to classical models, understanding the capacity of quantum neural networks is not well explored. Moreover, classical neural networks are known to produce highly degenerate Fisher information matrices, which can considerably slow down training. No such analysis has been performed for quantum neural networks.
- ✔ The feature map in a quantum model plays a large role in determining both its capacity and trainability via the effective dimension and Fisher information spectrum.
- ✔ Quantum Neural Network can be applied to various applications in reinforcement learning, communication networks etc
- ✔ There are challenges like Gradient Vanishing, Near-Term Device Compatibility and the possible Quantum Advantage.
- ✔ If one approaches simply based on this fact without the consideration of quantum gain, the result may be much inefficient compared to the existing classical algorithm. Therefore, in designing a new QNN-based deep learning algorithm, it is necessary to justify it by articulating its advantages over the corresponding classical models.[7]

Thank you